

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-09

BOTANY

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Topics To Covered

- 1** Factors Affecting Rate Of Photosynthesis
- 2** ...

Factors Affecting Rate Of Photosynthesis

- Two main factors:
 - a. Internal factors (plant factors).
 - b. External factors (environmental factors).

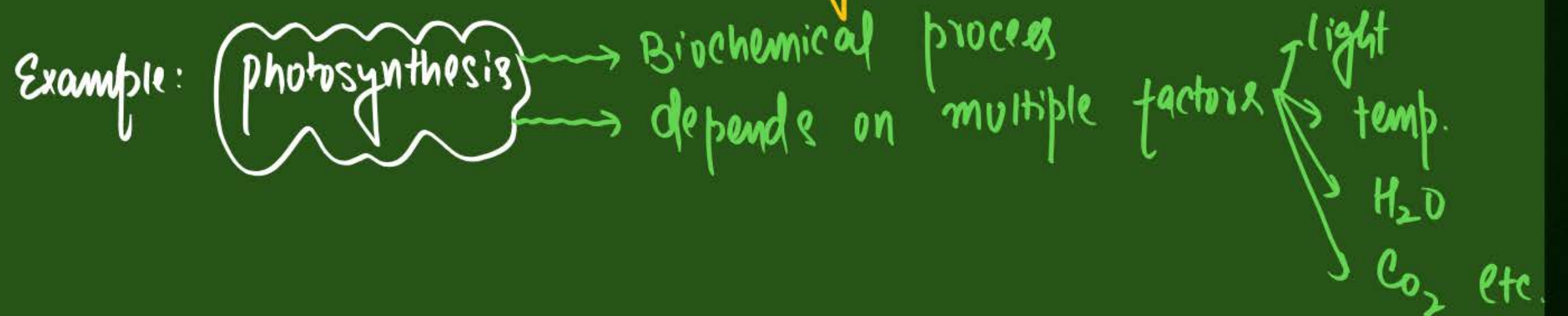
Internal Factors



- ①. Rate of photosynthesis $\propto$ No. of leaves.
- ②. Rate of photosynthesis $\propto$ Chlorophyll content of leaves.
- ③. Leaf Age
 - Immature leaf (Rate of photosynthesis low).
 - Mature / young leaf (Rate of photosynthesis high).
 - Old leaf (Rate of photosynthesis low / zero).
- ④. Rate of photosynthesis $\propto$ internal CO_2 Concentration (Because RuBisCo Enzyme binds with CO_2 and form Glucose)

Law of limiting factor

- Given by Blackmann (1905).
- AKA Blackmann's law of limiting factor.
- It states that: If a biochemical process is controlled by multiple factors, then its rate depends on the factor which is low/limiting.



External factors

① Light $\Rightarrow$

• It affects rate of photosynthesis by 03 different ways:

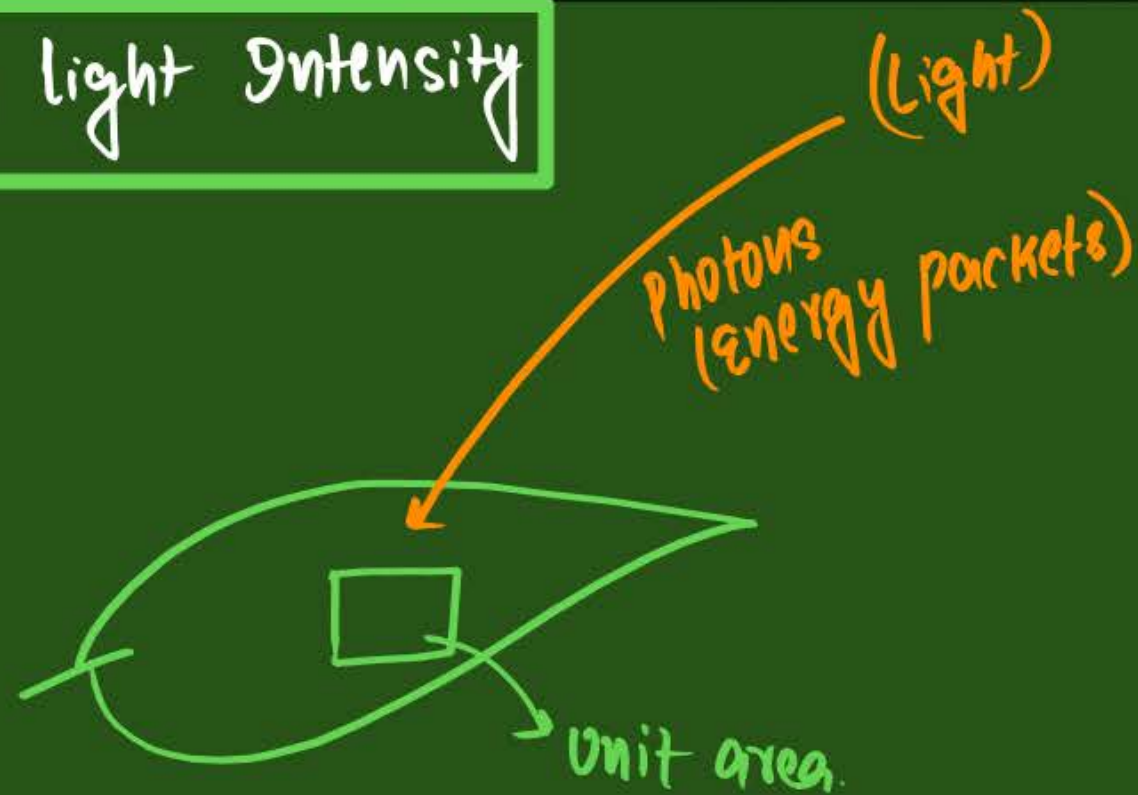
a. Light quality $\rightarrow$ Required quality for photosynthesis is PAR (400-700 nm)

b. Light duration $\rightarrow$ for how long light is required.

c. Light intensity $\rightarrow$ on detail.

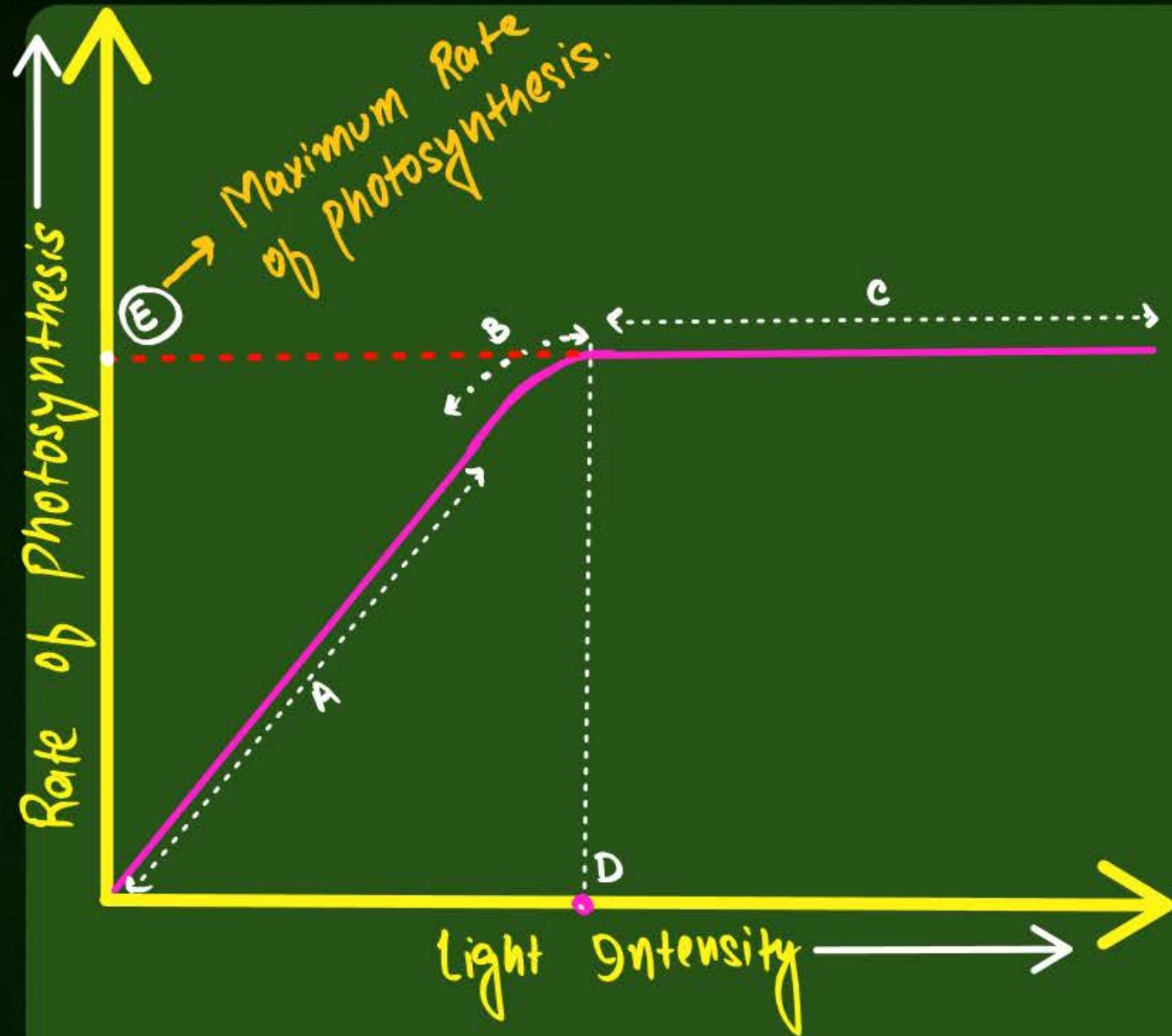
photosynthetically
Active Radiations.

Light Intensity



It is the number/amount of photons or energy packets (quanta) incident per unit area per unit time $\rightarrow$ Light Intensity

Light intensity is the most important factor which directly affects rate of photosynthesis.



① → In this region rate of photosynthesis is directly controlled by light intensity because in this region light intensity is limiting.

② → Gradual decrease in rate of photosynthesis.

③ → Light saturation point (that intensity of light at which plants show maximum rate of photosynthesis).

imp. → plants show saturation at 10% of total sunlight.

④ → In this region rate of photosynthesis is constant and does not depend on light intensity because here light intensity is not limiting.

NOTE

• In nature, light is rarely a limiting factor except for:

- a. plants under great depths in H_2O .
- b. plants growing canopy of large trees.
- c. plants growing under cloudy sky.
- d. plants growing in shade / dense forests.

②. Environmental CO_2 Concentration:

↳ It is major limiting factor for photosynthesis.

→ a) Atmospheric CO_2 Conc. = 0.03% to 0.04%.

(plants may tolerate max. 0.05% — beyond which plants may get damaged)

→ b) C_3 plants show Saturation at 0.045%.

→ c) C_4 plants show Saturation at 0.036%.

Conclusion: From (a), (b) and (c),

→ It is clear that atmosphere is providing sufficient CO_2 for C_4 plants and increase in atmospheric CO_2 concentration favours C_3 plants.

→ CO_2 becomes limiting for C_3 plants.

- ①. At low light intensity, neither C_3 nor C_4 plant respond to high CO_2 concentration $\longrightarrow$ because here light is limiting.
- ②. At high light intensity, both C_3 and C_4 plant show increase in rate of photosynthesis at high CO_2 concentration (but maximum benefit goes to C_3 plant).

CO_2 Fertilization Effect

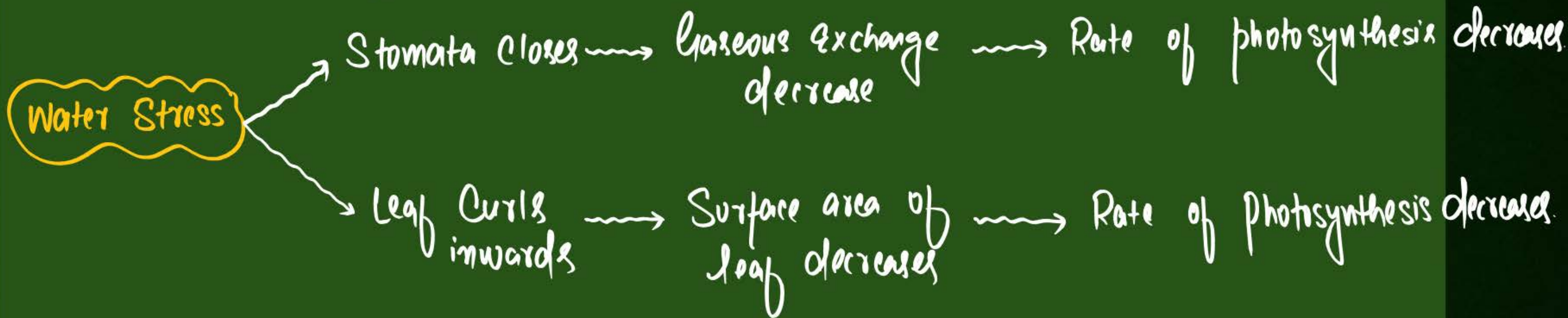
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 - It is increase in rate of photosynthesis by increasing CO_2 Conc.
 - It has been utilized for growing some green house crops like tomato, bell pepper etc. $\rightarrow \text{C}_3$ plants.
- These green house crops are allowed to grow in CO_2 enriched environments.

③ Temperature:

- Temp mainly affects Enzymes.
- Dark rxn. being highly Enzymatic are relatively affected by temperature as compared to light rxn.
- Optimum temp. for C_3 plants = $20-25^{\circ}C$.
- Optimum temp. for C_4 plants = $30-40^{\circ}C$.

④ Water:

- It is one of the reactants of light rxn.
- Water indirectly affects rate of photosynthesis.



Challenge for Week ②

Plant Kingdom

Morphology of flowering plants.

HOMEWORK



Revision

The graphic features the words 'Thank You' centered on a dark green background. 'Thank' is in yellow and 'You' is in white. The text is surrounded by a decorative border of red hearts. The hearts are arranged in a roughly circular pattern, with some hearts overlapping the text.

Thank
You